

Computer Graphics and Animation

BCA — Semester V — 2024 — Answer Sheet

Subject Code: CACS305

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. Differentiate between beam penetration method and shadow mask method. How does raster scan display system work?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. How DDA works? Digitize the line segment with the endpoints (15, 25) and (21, 35) using DDA line drawing algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What is a viewport? Consider a window with lower left corner at (2, 2) and upper right corner (5, 10) and a viewport with left lower corner at (3, 5) and upper right corner at (8, 8). What will be the value of the point in the viewport after the window to viewport transformation if the point is (4, 4) in the window?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Explain how scan line algorithm can be used for hidden surface removal.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Derive the transformation matrices for 3D rotation and reflections.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is animation? Explain about applications of Virtual Reality.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write short notes on (any two): a) RGB color model b) 2D shear transformation c) Computer Graphics Vs. Image Processing

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. Differentiate between boundary fill algorithm and flood fill algorithm in detail. Find the composite transformation matrix for anti-clockwise rotation of 60° about a point (2, 3). Use it to rotate a triangle ABC with vertices A (4, 3), (5, 5) and (8, 9).

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is visible surface detection? How object space method can be used for visible surface detection? Explain depth-sorting method for visible surface detection in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. How region codes are used in Cohen-Sutherland algorithm to clip the line segments? Find the clipping coordinates for lines PQ and RS with P(20, 5), Q(40, 30) and R(20, 30), S(80, 60) against the window with left upper corner(10, 50) and right lower corner(50, 10).

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.